Engineering Standard

23 January 2023

SAES-X-700

Cathodic Protection of Onshore Well Casing

Document Responsibility: Cathodic Protection Standards Committee

Table of Content

Summary of Changes

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 19 July 2020	**Current Revision** 23 January 2023		
1.2	1.2	Deletion	Galvanic anode CP for well casings
1.5	1.5	Modification	Install two pre-packaged 27.3 kg (60 lb) magnesium anodes inside the wellhead cellar for both coated and bare casings.
4.4	4.4	Modification	Max current capacity and number of well casings are not applicable for well pads where multiple well casings are installed within a same well pad
6.4.4	6.4.4	Modification	Bonding requirement for nearby structures other than well casings to 500 m
6.4.4	6.4.4	Modification	Air-cooled and oil-cooled rectifiers per 17-SAMSS-004 are permitted for well casing sites and these rectifiers shall be suitable for the location of installation.
Table 1A, 1B & 1C	Table 1A, 1B & 1C	Deletion	Separate Current Requirement for solar systems deleted and made same for both T/R and solar.
Table 1A, 1B & 1C	Table 1A, 1B & 1C	Addition	Requirement for flow lines and other pipelines that are made of Corrosion Resistance Alloys (CRAs) and or other high strength stainless steel materials like duplex stainless steel added
6.5.3	6.5.3	Addition	Steel casings for the active columns of these unconventional anode beds added to improve the anode

Paragraph Number		Change Type (Addition, Modification, Deletion, New)	Technical Change(s)
Previous Revision 19 July 2020	**Current Revision** 23 January 2023		
			bed performance in loss of circulation zones added
6.6.4	6.6.4	Modification	At the design stage only, the target commissioning current shall be achieved at a voltage of lower than 70% of the T/R rated voltage output.
6.11.2	6.11.2	Modification	Bonding requirement for nearby structures other than well casings to 500 m

1 Scope

1.1 This standard prescribes the minimum mandatory requirements governing the design and installation of new cathodic protection (CP) systems protecting new onshore metallic well casings. It may address existing wells if they are made part of a group containing new wells.

1.2 Cathodic protection systems for onshore well casings may be single dedicated, multiple or bonded ICCP CP systems.

Shared well casings are not allowed except for multiple well casings exceeding the limitations of multiple well casing CP system capacity (more than 5 wells, requiring higher than 150 Amp TR, etc.) as detailed in Section 6 below. In overlapping gas/oil fields, if a shared well CP system is not allowed. Single well casing CP system is not allowed where two wells are separated by less than 500 meters. Drilling islands can have more than 5 wells connected to the same rectifier.

1.3 New, existing, carbon steel, stainless steel, coated, bare, onshore oil, gas, unconventional gas, gas lift, observation, exploration, water injection, and water supply metallic well casings with a predicted life expectancy greater than five years shall be provided with cathodic protection within the time period specified in Section 7 of this standard if:

a. The well casing is installed through a corrosive formation, or

b. The well casing is not installed through a corrosive formation, but has a permanent buried metallic flow-line, in which case, sufficient CP shall be provided to maintain the flow-line at an acceptable protection level in accordance with SAES-X-400 without electrical isolation of the well casing from the flow-line.

Notes:

Typically, well casings without flow-lines completed in or above the UER formation do not require cathodic protection.

Where CP is not needed because the well casing is expected to last less than 5 years, if the proponent requires CP; then, it can be made out of scrap steel anodes.

1.4 A metallic well casing that is not installed through a corrosive formation and does not have a permanent flow-line does NOT require cathodic protection. However, if a foreign impressed current anode bed is within 500 meters radius of this type of well, electrical bonding shall be implemented to minimize the probability of downhole interference.

1.5 For oil and water wells only, a metallic well casing that is installed within a metallic or non-metallic cellar and backfilled shall be provided with galvanic magnesium anodes. Install two pre-packaged 27.3 kg (60 lb) magnesium anodes inside the wellhead cellar. Alternatively, clamp two 100-lb bare bracelet magnesium anodes within the top one meter of the well head and connect the cable to the well casing at the cellar zone to provide supplemental cathodic protection for the landing base area. When upgrading an existing CP system, depleted magnesium anodes at the cellar area shall also be restored. Measuring the potential and/or anode current output at the cellar area is not needed to meet any protection criteria, rather to ensure that the anodes are physically and electrically connected.

Note:

To provide further clarification for the above statement, supplemental galvanic anodes for permanent protection of the landing base area are not required for well casings that:

- *do not have cellars, or*
- *are installed in cellars that are not backfilled, or*
- *are FBE coated on the top three joints through the cellar area, or*
- *gas well casings.*

1.6 Replacements or in-kind replacements of existing, depleted or relocated anode beds and/or CP systems do not need CSD approval except if they are part of a new CP well installation. In such cases, existing wells do not need compliance with the commissioning requirements specified in this standard, however minimum monitoring current required shall be achieved as detailed in SAEP-333.

When a new well casing CP system is made part of a multi-well CP system hosting existing well casings, then, only if required upgrade the existing CP system to meet the monitoring current requirements of the existing wells.

Existing well casings without flowlines that lasted more than 30 years with no CP and no external corrosion leaks do not need a new CP system unless mandated by the proponent organization. The application of new CP in this case is not recommended as it could have adverse effects if gases resulting from the application of CP promoted the removal the corrosion products that prevented casing leaks. Exception is when CP systems are installed within 500 meters from unprotected structures.

The cathodic protection requirements for offshore well casings are beyond the scope of this standard and are addressed in SAES-X-300. Onshore well casings with anodes installed offshore (i.e., Pyramid anode) however are within the scope of this standard.

Cathodic protection as addressed in this standard does not provide corrosion protection for the inside surface of the well casing, tubing or liner.

Non-metallic (NM) flowline could act as an isolator hindering cathodic protection continuity between the well casing and trunkline; hence, connect the CP negative cable to the well casing rather than to the NM flowline. In such a case, only if needed then provide a dedicated CP system for the trunkline while ensuring adequate protection for all structures.

1.7 This standard shall not be attached to, nor made a part of a purchase order.

2 Conflicts and Deviations

Any conflicts between this document and other applicable Mandatory Saudi Aramco Engineering Requirements (MSAERs) shall be addressed to the EK&RD Manager.

Any deviation from the requirements herein shall follow internal company procedure SAEP-302.

3 References

All referenced specifications, standards, codes, drawings, and similar material are considered part of this Engineering Standard to the extent specified, applying the latest version, unless otherwise stated.

3.1 Saudi Aramco References

Saudi Aramco Engineering Procedures

SAEP-302	Waiver of a Mandatory Saudi Aramco Engineering Requirement
SAEP-332	Cathodic Protection Commissioning
SAEP-333	Cathodic Protection Monitoring

Saudi Aramco Materials System Specification

17-SAMSS-004	Conventional Rectifiers for Cathodic Protection
17-SAMSS-006	Galvanic Anodes for Cathodic Protection
17-SAMSS-007	Impressed Current Anodes for Cathodic Protection
17-SAMSS-008	Junction Boxes for Cathodic Protection
17-SAMSS-012	Photovoltaic Power Supply for Cathodic Protection
17-SAMSS-017	Impressed Current Cathodic Protection Cables
17-SAMSS-018	Remote Monitoring Units (RMUs) for Cathodic Protection

Saudi Aramco Engineering Standards

SAES-B-062	Onshore Wellsite Safety
SAES-P-104	Wiring Methods and Materials
SAES-X-300	Cathodic Protection of Marine Structures
SAES-X-400	Cathodic Protection of Buried Pipelines

Saudi Aramco Best Practices

SABP-X-003	Cathodic Protection Installation Requirements

Saudi Aramco Standard Drawings

AA-036385	Cathodic Protection - Deep Anode Bed
AA-036389	Galvanic Anode Details
AD-036785	Symbols for Cathodic Protection

Saudi Aramco General Instructions

GI-0002.710	Mechanical Completion and Performance Acceptance of Facilities

3.2 Other References

National Fire Protection Association

NFPA 70	National Electrical Code (NEC)

The Association for Materials Protection and Performance (AMPP/NACE)

NACE SP0186-2007	Application of Cathodic Protection for External Surfaces of Steel Well Casings

4 Terminology

4.1 Acronyms

AOC	Aramco Overseas Company, The Hague
ASC	Aramco Services Company, Houston
ANSI	American National Standards Institute
ASTM	American Society for Testing and Materials
AWS	American Welding Society
ASME	American Society of Mechanical Engineers (USA)
CSA	Canadian Standard Association
CSD	The Saudi Aramco Consulting Services Department
FM	Factory Mutual Research Corp.
GOSP	Gas and Oil Separation Plant
JEDEC	Joint Electronic Devices Engineering Council

MSAER	Mandatory Saudi Aramco Engineering Requirements
NEC	National Electrical Code
NEMA	National Electrical Manufacturers Association (USA)
RSA	Saudi Aramco Responsible Standardization Agency
SAES	Saudi Aramco Engineering Standard
SAMSS	Saudi Aramco Materials System Specification
SI	International System of Units
SSPC	Steel Structures Painting Council (USA)
UL	Underwriter's Laboratories, Inc. (USA)

4.2 Definitions

The following general list of definitions and abbreviations has been developed specifically for application with Saudi Aramco cathodic protection standards and specifications:

AA:	The ANSI cooling class for a dry-type self-cooled transformer or reactor that is cooled by the natural circulation of air. Syn.: AN (IEC), see definition 3.1 in IEEE C57.12.
Anode:	An impressed current or a galvanic anode for cathodic protection applications.
Anode Cable:	A cable directly connected to an impressed current or galvanic anode.
ANV:	The ANSI cooling class for a dry-type non-ventilated self-cooled transformer, which is so constructed as to provide no intentional circulation of external air through the transformer, and that operates at zero gauge pressure. Syn.: ANAN (IEC), see definition 3.15 in IEEE C57.12.
Armored Cable:	Armored cable for cathodic protection is a single core insulated cable manufactured with a double layer spiral wound insulated galvanized steel sheath to provide mechanical protection, and typically used for subsea applications.
Bond Cable:	A cable installed between two metallic structures to provide electrical continuity between the structures for the purpose of cathodic protection.
Buyer:	Saudi Aramco Purchasing Department Representative.
Buyer's Representative:	The person or persons designated by the Purchasing Department to monitor/enforce the contract. Normally, this is the on-site inspector.
Calcined Petroleum Coke Breeze:	A carbonaceous backfill used as a conductive backfill media for impressed current anodes in soil.

Christmas Tree: The set of valves, spools and fittings connected to the top of a well to direct and control the flow of formation fluids from the well.

Coated Casing: The term "coated casing" as used in this engineering standard describes a well casing with an external non-conductive coating (typically Fusion Bonded Epoxy or FBE). The coating must be applied to all sections of the casing in contact with soil or formation, from surface to the bottom of the casing or to a depth determined to facilitate external corrosion mitigation with cathodic protection through the relevant down hole corrosive aquifers. Casings that have been coated over the upper two or three joints of casing only are not "coated casings". Coating applied to well casings is not applied as a corrosion barrier. It is applied to reduce the total amount of CP required or to extend the influence of the applied CP.

CP: Cathodic Protection

CP Assessment Probe: A multi-electrode probe custom fabricated for Saudi Aramco that is designed to facilitate representative measurements of instant off, total polarization, current density, and resistivity.

CP Coupon: A cathodic protection coupon used to measure representative potentials or current densities on a pipeline or other buried or immersed metallic structure.

CP System Operating Circuit Resistance: The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.

CP System Operating Circuit Resistance: The total resistance seen at the output of a CP power supply or the total working resistance in a galvanic anode system.

CP System Rated Circuit Resistance: The rated output voltage of a cathodic protection power supply divided by the rated output current. For photovoltaic power supplies, the rated output current is the design commissioning current.

Cross Country Pipeline: A pipeline between; two plant areas, another cross country pipeline and a plant area, or between two cross country pipelines.

Deep Anode Bed: Anode or anodes connected to a common CP power supply and installed in a vertical hole with a depth exceeding 15 m (50 ft).

Design Agency: The organization responsible for the design of the pipeline and associated CP system. The Design Agency may be the Design Contractor, the Lump Sum Turn Key Contractor or an in house design organization of Saudi Aramco.

Flowline: A pipeline connected to a well.

Flare Line or Blow Down Line: A line for pumping out of unwanted gas or hydrocarbon

Foundry (for anodes): An anode foundry is a facility that produces metal castings from either ferrous or non-ferrous metals or alloys.

Galvanic Anodes: Anodes fabricated from materials such as aluminum, magnesium or zinc that are connected directly to the buried structure to provide cathodic protection current without the requirement for an external cathodic protection power supply. Galvanic anodes are also referred to as sacrificial anodes.

Hazardous Areas: An area where fire or explosion hazards may exist due to flammable gases or vapors, flammable liquids, combustible dust, or ignitable fibers or filings (see NEC Article 500).

HDD: Horizontal Directional Drilling

ICCP: Impressed Current Cathodic Protection

Impressed Current Anodes: Anodes typically fabricated from High Silicon Cast Iron (HSCI), Mixed Metal Oxide (MMO) or liner anode, etc. that are connected through a DC power supply to the buried or immersed structure to provide cathodic protection current.

Kill Line: A high-pressure pipe leading from an outlet on the BOP stack to the high-pressure rig pumps

Lead Wire (for anodes): A cable directly connected to an anode.

Manufacturer: The Company that assembles the components for the finished product and provides the finished product either through a Vendor or directly to Saudi Aramco.

Resistivity Meter (Megger): A meter designed to measure ground resistivity, or can be connected to measure resistance in a format that excludes the resistance of the test wires.

Multiple Well (Multi-well) Casing CP System: Two or more well casings protected by one common CP rectifier system.

Mill (for anodes): An anode mill is a facility that produces anodes such as MMO or Platinized Niobium (without lead wires) for cathodic protection applications (or any other anode type that does not involve casting).

Negative Cable: A cable that is electrically connected (directly or indirectly) to the negative output terminal of a cathodic protection power supply or to a galvanic anode. This includes bond cables to a cathodically protected structure.

Non-metallic Flowline: A flowline that is made out of non-metallic materials.

Non-incendive Equipment: Equipment having electrical/electronic circuitry that is incapable, under normal operating conditions, of causing ignition of a specified flammable gas-air, vapor-air, or dust-air mixture due to arcing or thermal means. (See NEC Article 100)

Off-Plot: Off-plot refers to any area outside of the plot limits.

ONAN: The ANSI cooling class for a transformer or reactor having its core and coils immersed in mineral oil or synthetic insulating liquid with a fire point less than or equal to 300°C, the cooling being effected by the natural circulation of air over the cooling surface. (ONAN was previously termed OA). See definition 3.303 in IEEE C57.12

On-Plot: On-plot refers to any area inside the plot limit.

Perimeter Fence: The fence which completely surrounds an area designated by Saudi Aramco for a distinct function.

Photovoltaic Module: A number of solar cells wired and sealed into an environmentally protected assembly.

Pipeline: The term "pipeline" is used generically to refer to any type of pipeline.

Plant Area: The area within the plot limits of a process or storage facility.

Plot Limit: The plot limit is the boundary around a plant or process facility. The plot limit may be physical such as a fence, a wall, the edge of a road or pipe rack, chains and posts or a boundary indicated on an approved plot plan.

Positive Cable: A cable that is electrically connected (directly or indirectly) to the positive output terminal of an impressed current cathodic protection power supply, including impressed current anode cables.

Process Pipeline: A pipeline typically associated with a plant process and typically above ground within a plant facility.

Production Pipeline: A pipeline transporting oil, gas or water to or from a well. These include flowlines, test lines, water injection lines, and trunklines.

Pyramid Anode: An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and placed on the sea bed. The pyramid anode derives its name from the pyramid shaped concrete base.

Reference Electrode: An industry standardized electrode used as a common reference potential for cathodic protection measurements. A copper/copper sulfate (Cu/CuSO4) reference electrode is typically used for soil applications. A silver/silver chloride (Ag/AgCl/0.6M Cl) reference electrode is typically used for aqueous applications.

Responsible Standardization Agent (RSA): For CP material, this would usually be the Supervisor of the CSD Cathodic Protection Team or the Saudi Aramco CSD cathodic protection Subject Matter Expert.

Single Well Casing CP System: One well casing CP system protected by one CP rectifier system.

Soil Transition Point: The surface location where a pipeline enters or exits the soil, i.e., above grade to below grade transition, or below grade to above grade transition.

Subject Matter Expert (SME): The SME is the assigned Consulting Services Department cathodic protection specialist.

Shared Well Casing CP System: Two or more well casings protected by two or more CP rectifier systems that are negatively bonded thru a Dedicated Bonding Facility arrangement. Shared well casings are not allowed except for solar systems.

Surface Anode Bed: Anode or anodes connected to a common CP power supply, installed either vertically or horizontally at a depth of less than 15 m (50 ft).

Tension Spring Anode: An offshore anode assembly custom manufactured for Saudi Aramco that is fabricated with mixed metal oxide anode components and suspended on a wire rope/tension spring assembly beneath an offshore platform.

Test Line: A pipeline that is used for testing an individual well or group of wells.

Thermite Weld: An exothermic process for use in making electrical connections between two pieces of copper or between copper and steel.

Transformer/Rectifier (T/R): A CP power supply that transforms and converts AC to DC power, often referred to in the cathodic protection industry as a rectifier.

Transmission Pipeline: A cross country pipeline transporting product between GOSPs WIPs or other process facilities.

Trunkline: A pipeline designed to distribute or gather product from two or more wells, typically connecting flowlines or injection lines to the associated GOSP or WIP.

Utility line: A pipeline that delivers a service product (typically water, gas or air).

Unconventional Deep Anode Bed: Anode bed drilled to typical depth of 120 meters thru dry drilling using air or foam with a geotextile liner to 120 meters and PVC water reservoirs on the top 60 meters.

Vendor: A company that receives a purchase order to supply the finished product, material or equipment. The Vendor may also be the Manufacturer.

Venturi Spool: A gas metering spool recording apparatus at gas well sites.

WIP: Water Injection Plant

Well Casing: Large-diameter pipe lowered into an open hole and cemented in place.

Well Head: The system of spools, valves, and assorted adapters that provide pressure control of a production well.

5 Design Review and Approval

5.1 Contractor/Designer Qualifications

5.1.1 Cathodic protection designs shall be completed by CSD approved local design offices with a minimum of five years verifiable cathodic protection design experience and a minimum industry qualification of NACE CP Level 4 and BS degree. Indicate the NACE CP level, certification number, name of designer and/or his signature/initial in the design package.

5.1.2 Field measurements required for the design, survey, installation, walk-through, pre-commissioning, commissioning, inspection, monitoring, restoration, replacement, troubleshooting, field-investigation, assessment, drill stem analysis, anode bed installation, etc. shall be performed by an Engineer or Technician with a minimum industry certification level of NACE CP Level 2. Indicate the NACE CP level, certification number, name of designer and/or his signature/initial in the design package.

Note:

PMT may request assistance from CSD for the verification of the qualifications of the Design Contractor's engineer responsible for designing the CP systems and to approve new designers.

5.2 Design Review

5.2.1 DBSP and project proposal does not require conducting a site visit unless that is mandatory to complete the package. The proposed construction drawings and the related cathodic protection design information for every design package shall at minimum be submitted to the CP Proponent organization and to Saudi Aramco's Consulting Services Department (CSD) or where applicable to the CSD approved local design office Engineers for review and approval. The most up to date list of approved design offices is posted on ShareK.

5.2.2 The Design Agency shall not issue drawings for construction until the design has been reviewed and approved in writing by CSD or where applicable by the CSD approved local design office Engineers and the CP Proponent organization.

5.3 Design Basis Scoping Paper (DBSP)

Unless specifically requested by PMT, DBSP does not need to be reviewed or approved by CSD per SAEP-303. The DBSP shall specifically state if cathodic protection is required, or is not required, as dictated by SAES-X-700. No other design considerations for cathodic protection are required for the DBSP.

5.4 Project Proposal

5.4.1 Unless specifically requested by PMT, project proposal does not need to be reviewed or approved by CSD Project Proposal packages submitted to CSD for review shall provide all design considerations that can be developed without requiring the measurement of field data or a site visit.

5.4.2 The Project Proposal package shall include a specific statement in the scope of work that clearly identifies any requirement to provide CP for existing well casings, flow-lines or trunk-lines.

5.4.3 The Project Proposal shall provide clear direction on the general design approach with respect to the following:

a. The CP systems shall be designed as:
   - single well CP systems, or
   - multi-well CP systems

b. The well casings are:
   - coated casings, or
   - bare casings

c. The CP power supply shall be:
   - Photovoltaic, or
   - AC powered
   - air cooled, or
   - oil cooled
   - single phase, or
   - three phase

5.4.4 The Project Proposal shall clearly state if remote monitoring equipment will be included per 17-SAMSS-018 with the CP power supplies and if so it shall provide the following general design information:

a. Will the CP power supply be provided with:
   - signal transmitters, or
   - a Remote Monitoring Unit (RMU)

b. What operating parameters are going to be monitored (must meet minimum requirements)

c. What communication system will be used for the remote monitoring system?

d. What software will be supplied for the CP Proponent? If the data is not transmitted to the area Saudi Aramco PI data base; then, the software shall be provided through the Project.

5.4.5 The Project Proposal shall contain a professionally drafted Index “X” CP layout drawing using the cathodic protection symbols shown on Standard Drawing AD-036785 “Symbols for Cathodic Protection”, illustrating:

a. All existing and new well casings, buried/above grade metallic/non-metallic, shorted/isolated flow-lines, test lines, pipelines, blowdown lines, flare lines, bypass lines and trunk-lines associated with or affected by the proposed CP system within the 1,000 meter radius of the new well casing.

b. The proposed location of all CP equipment associated with the new CP system with general details for:

- the proposed anodes and anode bed(s), location, type (surface, deep conventional, deep unconventional), depth, number of anodes, anode bed derating, anode to anode spacing, anode bed to anode bed spacing, anode bed to buried structures spacing
- output ratings and condition for the existing/proposed CP power supply and available spare capacity
- cable locations/route, lengths, number, and sizes
- underground/above grade, metallic/non-metallic, existing/new junction boxes, splice boxes, and bond stations and their condition.

5.5 Detailed Design

5.5.1 Detailed Design packages shall provide all design considerations contained in the Project Proposal further developed based on the collection of relevant field data including:

a. Measured soil resistivity for proposed anode beds that will be installed shallower than 15 meters (surface) or deeper than 15 meters (deep)

b. Site information sufficient to develop a field verified site plan (CP layout drawing) that clearly illustrates all proposed CP equipment and all buried structures within 1,000 meter radius of the new well casing(s) location.

c. Field verified operating data for existing CP equipment within 1,000 meter radius of the new well casing(s) location.

d. For detailed designs of multi-well CP systems in fields where the casing resistances have not already been determined, site tests on each relevant type of well casing to determine the respective resistance-to–ground and structure back emf are required.

5.5.2 Detailed Design packages shall contain the **original** completed and signed by CSD or where applicable by the company approved local design offices “Well Casing Design Quality Assurance Check List” attached to the cover letter and/or transmittal sheet for the design package (See Appendix 1).

5.5.3 Detailed Design packages shall contain all calculations and applicable field data required to verify design compliance with the Saudi Aramco Cathodic Protection Engineering Standards including an electrical simulation drawing for all multi-well and shared-well CP systems. Shared well casings are not allowed except as detailed in Section 6 below.

6 Design Technical Requirements

6.1 General

Company approved CP designers are allowed to conduct the following activities in lieu of specific review by Saudi Aramco:

1) Develop the 90% detailed design CP package and IFC for single and multiple well casings. While single & Single with pipeline bonding shall go to construction without Saudi Aramco review or approval, multiple still requires Saudi Aramco approval.

2) Analyze the drill stem and test anode resistances and provide anode installation recommendations.

3) Evaluate and approve the pre-commissioning report for single oil and water well casings.

6.1.1 Shared well casings are not allowed except for solar systems especially bare and for multiple well casings exceeding the limitations of multiple well casing CP system capacity (more than 5 wells and/or requiring higher than 150 Amp TR, etc.). In overlapping gas/oil fields a shared well casing is not allowed and thus it is a clear violation of this standard.

If a shared well CP system is installed (although not allowed by this standard), then it will not be approved but rather becomes the sole and full responsibility of the proponent organization who approved/requested that, including the risk for well casing leak due to possible CP interference.

Shared well casings are only allowed in the following conditions provided they are pre-approved by the proponent organization(s) in case of overlapping gas/oil fields:

1) more than 5 wells within 500 meters from each other or from the most recent well

2) current requirement is larger than 150 Amps

3) when the power source is solar especially for bare well casings

4) when the two CP systems are solar.

In overlapping fields that involve oil & gas wells with multiple well CP system, the responsibility for the CP system is that of proponent of most recently installed well. That is if the most recently installed well is gas then, gas proponent is responsible for the CP system monitoring & maintenance. Likewise, if oil is the most recently well then, the oil proponent is the responsible organization.

The proponent organization is responsible for connecting the CP system after drilling and workover completed their drilling or wire line operations.

6.1.2 The design shall facilitate an integrated CP system for all associated buried metallic structures, and shall comply with all spacing and access restrictions detailed in SAES-B-062.

6.1.3 CP power sources for an existing CP system shall not be utilized or be made part of a multiple system to protect a new well casing within 500 meters radius without a prior approval from the CP Proponent organization for the existing CP system. PMT is simultaneously responsible to upgrade the existing CP system to meet the commissioning requirements of the new well(s) and the monitoring requirements of the existing well(s) and to meet the monitoring potentials of all affected pipelines/flowlines within 1,000 meters from the new well casing. At gas/oil overlapping fields, PMT is responsible to obtain a prior approval from both CP Proponent organizations of the overlapping fields about the plan to use single, multiple CP system (shared is not allowed).

6.1.4 Permanent stainless steel, carbon steel, bare, coated buried flare lines and blow down lines on a well pad do not need a dedicated CP system, but rather they shall be made permanently electrically continuous with the well casing or the negative circuit of the well casing CP system to ensure their adequate protection. If must be isolated, then protect them with galvanic anode system. Bond across Venturi spools with a metallic strap at gas wells to ensure electrical continuity with the CP system and to minimize the possibility of CP interference. Where non-metallic pipelines are used (spools, flowlines, test lines and truck lines) bonding across using a dedicated bond box is required. If isolation is a must, then provide adequate CP for all metallic parts.

6.1.5 The use of a photovoltaic CP power supply shall be assessed by PMT from an economic perspective for cathodic protection of new wells or new fields considering the nearest 4.16 kV or 13.8 kV power line.

Notes:

The economic assessment shall include comparative costs for operations and maintenance between the systems being considered. During the DBSP review, the proponent organization can then verify stipulated versus actual costs, and may consider other cost related factors like future expansion and theft/vandalism issues.

For designs where photovoltaic power supplies are determined to be the most cost effective alternative, consideration should be given by PMT to place a request with the Saudi Aramco Drilling Dept. to have the well casings externally coated to reduce the CP current requirement. Photovoltaic CP for a coated well casing typically has a notable cost advantage over a bare well casing.

6.1.6 When demonstrated by the simulation program with calculated resistance and current requirements, a dedicated CP power source shall be provided for each well when the surface facilities for the two most distant wells involved are separated by more than 500 meters. Otherwise, use multiple design if the distance is within or less than 500 meters.

Notes:

In areas where theft and vandalism of a solar powered CP system and cables is a concern to the CP Proponent, then to avoid theft of long runs of buried cable, the cable shall be installed with approved below grade and/or non-metallic junction boxes buried concrete anchors at each cable end, and with minimal use of above grade cable identification markers.

6.1.7 Well casings separated by less than 500 meters shall utilize a multiple CP power source, provided the design protection criteria for each well as stated in Tables 1A, 1B, and 1C of this standard are met without the use of electrical resistors at the design stage. Electrical resistors maybe used at the commissioning and/or operating stages provided that they are intrinsically safe and are not installed in hazardous zones.

Exception:

If during commissioning of multi-well CP system, adequate current distribution is not achieved, then resistors may be used. Resistors shall be welded tap adjustable or fixed (non-adjustable) and shall be 0.15 ohms or less.

Note:

Coated well casings with normally above grade flowlines may use the flowline for negative cable connection and hence do not require a dedicated negative cable to each well but are still restricted by the 500 meter separation imposed on CP systems with multiple well casings and shall meet all well casing and flowline protection requirements.

6.1.8 At sites where one CP power source is used to protect multiple well casings, each uncoated well shall have a dedicated negative cable connection. The negative cables shall be terminated in a negative junction box located to optimize the current distribution between casings.

6.1.9 Approved underground junction box may be used for an anode bed junction box, bond box, splice box, or a negative cable junction box.

Note:

Underground junction boxes restrict the ability to measure current distribution to the anodes or the structures and are only recommended for use in areas where known repetitive cable theft and vandalism or other operating conditions prohibit the use of above ground junction boxes.

6.2 Field Data

Field data and site verification of existing equipment that may affect or be affected by the proposed CP system are required for the detailed design.

The following field investigation and data collection shall be completed and the results included with all detailed design packages submitted for approval.

6.2.1 Existing Well Casings and Buried Pipelines

Verification and documentation (illustrate on the CP layout drawing) of all buried pipelines and well casings, or other buried metallic structures within 500 meters from the new well casing and the location of the proposed CP power supply, anode bed, junction box, fence, rectifier, pipeline, flow line or well casing.

6.2.2 Existing CP Systems

Measurement of all relevant cathodic protection currents in cables, junction boxes, anode beds, grounding systems and flow-lines, plus all relevant power supply voltages and currents, and all relevant structure potentials. PMT is required to measure the well casing current at the well head below the negative cable connection point especially for gas wells using a clamp on ammeter. In addition, use the clamp to re-confirm the well casing net current by measuring the current at the flowline, kill line, test line, blowdown line, annulus line, trunk line, flare line, grounding and all other connection points to the well head. Measurement point on the wellhead P&ID and grounding drawings shall be used to identify all underground grounding components, the type of grounding, the way it is connected to the CP system and or site structures, whether isolated or shorted in order to account for the grounding current requirement at the design stage. Designs that neglect grounding shall be rejected. Installations that do not meet all commissioning requirements shall be re-sent to the design office for a re-design and simulation if required. Saudi Aramco can be consulted to resolve non-routine and complex issues, especially for multiple well casings. All CP pre-commissioning requirements for new wells and flowlines shall be met and all monitoring requirements for existing wells and flowlines shall also be met.

6.2.3 Soil Resistivity Measurement

Soil resistivity measurements are mandatory for all types of new and existing anode beds; assumed or back calculated resistivities are not allowed.

6.2.3.1 Shallow (Surface) Remote Anode Beds

a. Even after conducting surface resistivity measurements, the preference is to use deep anode beds unless the site conditions permit the use of surface anode beds, i.e., Subkha soils. Install a watering system in very dry soil or use a deep anode bed. Surface soil resistivity or soil conductivity measurements shall be taken at 10 meter intervals. Soil conductivities maybe measured using the four-pin, three-pin or non-contact electromagnetic tools.

b. Measurements shall be taken for at least two soil layers (typically 3 and 6 meter depths) down to the target installation depth of the anode bed. Calculate the layer resistivity to be used in the design.

c. Measurements shall be taken in parallel and perpendicular orientations over the full length of the proposed shallow anode bed location.

d. If the soil resistivities within a proposed shallow anode bed vary by more than 100%, either additional anodes shall be provided or, anodes of the same composition with a higher current capacity can be placed in the low resistivity areas so that no anode exceeds the maximum commission current (Table 3).

e. The four-pin Wenner method or a non-contact electromagnetic soil resistivity/conductivity instrument shall be used as appropriate for the location where soil resistivity is measured. Type of measurement and the equipment used shall be based on the effectiveness and reliability of the selected method and equipment capability.

f. The proposed location for the shallow anode bed shall be clearly marked with wooden or metal stakes at each end. Show the location on the CP layout drawing.

6.2.3.2 Deep Remote Anode Beds

a. Unless mandated and approved by the proponent, the max deep anode bed depth is 120 meters, max no. of anodes per hole is 20 anodes, anode to anode spacing shall be between 3 and ½ a meter, anode hole to anode hole spacing shall be within 10 to 20 meters and use of slotted/non-slotted PVC or metallic casing is not allowed, except for un-conventional deep anode bed design.

b. Use un-conventional deep anode bed systems in areas where the measured soil resistivity is extremely high (typically higher than 50,000 ohm-cm) and/or the strata is known to have cracks, voids, gaps or multiple loss circulation zones. Soil resistivity or soil conductivity measurements for both conventional and un-conventional deep anode beds are mandatory for new/existing well casing sites and/or new CP system upgrades.

c. Only use Saudi Aramco recommended/approved drilling contractors for drilling anode beds. Record drill stem and test anode resistance measurements on the form contained in Appendix 2 and submit for review and analysis to Saudi Aramco or to approved local design offices as applicable.
Drill stem and test anode measurements are not required for unconventional deep anode holes. The drill stem and test anode measurements shall be taken in accordance with the requirements detailed in Standard Drawing AA-036385. Saudi Aramco or approved local design offices as applicable shall determine the final acceptable borehole depth, number of anodes, anode to anode spacing, number of anode beds, separation between anode beds, size of the rectifier, use of resistors and anode distribution.
Water level in the anode hole must be at the top of the casing before taking each measurement.

d. Where a conventional deep anode bed was used but did not meet the requirements due to high resistance, drill the new anode beds as unconventional with a prior approval from the proponent.

e. The proposed location for the deep anode bed shall be clearly marked with a wooden or metal stake. Show the location on the CP layout drawing.

f. While drilling, PMT shall hold the job if any of the following is encountered: high anode cable resistances, loose connection to resistance meter or loss of fluid circulation during drilling, coke breeze pouring, or anode installation.

g. If commissioning could not be met due to facing high anode bed resistance, then it could be due to missing drilling procedures, wrong equipment or consumables, extremely high resistivity soils, improper installation practice, backfill settlement, air gaps, anode-cable contact issues, dry backfill, loss of circulation fluid, cavities, missing intermittent check/hold points. Possible solutions include, to prepare a clear procedure for anode bed installation with check and hold points, verify anode to cable resistance before lowering the anodes, verify during backfilling withhold points. Water volume, calcined petroleum coke volume required shall be calculated in advance and full calculated quantity must be filled and recorded. Qualify and approve the drilling contractors to meet to the procedure. Sample calcined petroleum coke to be lab tested for its quality. Investigate foam drilling/steel casings for loss of water areas.

h. When facing erratic drill stem and test anode data then that is probably due to poor contacts, incorrect measurement practices or dry measurements. Potential solutions could be to use qualified persons for measurements, always ensure proper electrolytic continuity, maintain wet conditions, check water levels, and if necessary add water, be able to check and re-measure if required before sending the data for further processing, temporary power up the anode bed following anode installation for current distribution check.

6.2.3.3 Distributed Anode Beds

Distributed anodes are not used for well casing CP.

6.2.3.4 Galvanic Anode Locations

a. Galvanic anodes are not used for well casing protection. Galvanic anodes when used for protection of flowlines or other nearby pipelines, soil resistivity measurements are. Soil resistivity measurements are not required for galvanic anodes used for supplemental cathodic protection of associated pipelines.

b. Soil resistivity measurements taken by the design agent, and ground water resistivity data supplied by the Saudi Aramco Groundwater Protection organization, shall be used for the design of galvanic anode systems for pipelines.

6.3 Design Life

6.3.1 Size ICCP anode beds to discharge the CP power source rated current at the anode consumption rate detailed in Table 2, for a minimum of 20 years for HSCI anodes. MMO anodes shall be designed for 25 years.

$$\left[\frac{\text{Total Weight of all Anodes (kgs.)}}{\text{Anode Consumption Rate x CP Power Source Design Current Capacity}}\right] \geq 20 \text{ Years}$$

6.3.2 For flowlines, galvanic anode systems used for temporary or supplemental CP shall be designed to provide a minimum life of 10 years. Galvanic anode systems used for permanent CP shall be designed to provide a minimum life of 20 years for flowline CP.

6.3.3 New Well Added to a CP System Less than 10 Years Old within 500 meters:

Use multiple CP systems for the addition of a new well within 500 meters from an existing well(s), up to 5 wells and up to 150 Amp rectifier rating. Otherwise PMT after securing the proponent approval may use a shared CP system when in compliance with the exceptions for permitting the use of shared CP systems as detailed in Section 6 above. Upgrade the rectifier to a higher level if needed and ensure that all new and existing well casings within 500 m and flowlines within 1,000 are adequately protected. While new structures shall meet the commissioning requirement, existing structures shall meet the monitoring requirements.

Where a new well is being added to a CP system commissioned within the previous 10 years, an anode bed upgrade is NOT required if the existing T/R and anode bed are capable of providing the required current at an output voltage not greater than 90% (±10%) of the power supply full rated voltage. If the required current for the existing wells and the new well(s) cannot be achieved at 90% (±10%) or less of full rated voltage, an anode bed upgrade shall be designed to comply on a prorated basis (see 6.3.4). Upgrade the rectifier to a higher level if needed.

6.3.4 New Well Added to a CP System more than 10 Years Old within 500 meters

If requested by the proponent, evaluate replacing the rectifier, junction box, bond box and splice box if more than 20 years old. Where a new well is being added to a CP system that is more than 10 years old, the contribution of the existing anode bed to the overall anode bed life and current capacity for the upgraded CP system shall be calculated on a prorated basis using 10 years at full rated output as full life. The existing anode bed shall be bonded to the new anode bed if cost-optimal and in compliance with this standard. Use of the existing anode beds should also be evaluated economically and practically as they could be too far away and thus might be cost prohibiting or impractical to utilize. Upgrade the rectifier to a higher level when needed. Use of existing facilities requires the proponent's approval.

Notes:

Example: 12 TA-4 anodes operating for 5 years at 23 amps would be given a prorated remaining life contribution calculated as follows:

- *1 TA-4 anode has a capacity of 4.45 Amps x 10 year = 44.5 Amp-yr*
- *12 TA-4 have a 10 year capacity of 44.5 x 12 = 534 Amp-yr*
- *12 TA-4 at 23 Amps for 5 years have consumed 5 x 23 = 115 Amp-yr*
- *Prorated percent consumed = 115/534 x 100 = 21.5%*
- *Remaining prorated capacity = 78.5% x 12 anodes = 9.4 = 9 anodes*

6.4 Design Current Criteria

6.4.1 Use externally coated casings for all new, onshore, oil, gas, unconventional gas, gas lift, observation, exploration, water injection, and water supply metallic wells regardless of the well depth. This will minimize the CP current requirement, allow use of multiple well casings, reduce AC consumption and permit deployment of solar systems at remote areas. If a bare well casing is used, then it shall be pre-approved by the proponent and shall not be violating the general decision chart of coating in Appendix A3.3 of this standard. The cathodic protection system design shall provide the minimum design currents detailed in Tables 1A, 1B, and 1C of this standard.

6.4.2 The CP requirements for well casings in fields specifically covered by Tables 1A, 1B, or 1C have resulted from actual Corrosion Protection Evaluation Tool (CPET) logging. Other fields were not logged and thus should be determined by future logging or extrapolation. Rectifier sizes in Tables 1A, 1B, or 1C may be used as a guideline while alternative cost-optimum sizes are also allowed. Upgrade the rectifier to a higher level whenever needed. A list of recommended CP rectifier rating is provided in Table-6 for reference.

6.4.3 The protection criteria for commissioning and monitoring are detailed in Tables 1A, 1B, and 1C of this standard. Tolerance values of ±10% on commissioning values during commissioning and on monitoring values listed in Tables 1A, 1B, and 1C of this standard shall be implemented where needed.

6.4.4 **Multi-well (multiple)** CP systems for new well(s) located within 500 meters from the nearest well shall be designed with one CP power supply sized with a minimum rated output current equivalent to the sum of the commissioning current requirement plus the estimated flow-line, test line, kill line, flare line, blow down line and grounding system current requirement plus a design surplus of 20%. Tolerance of ±10% may be considered when sizing the rectifier to select a standard rectifier rating or to use the existing CP system. Max rectifier size shall be 150 Amp, max rectifier voltage shall be 100 V and max number of well casings per rectifier shall be 5 wells (Max current capacity and number of well casings are not applicable for well pads where multiple well casings are installed within a same well pad). The 20% surplus current is not required for single well CP system. Upgrade the rectifier to a higher level whenever needed. Ensure adequate protection for all buried pipelines and flowlines that belong to the new well or to other facilities if they fall within 500 meters from the new well casing. Carryout field survey and provide a dedicated bonding with a bond box when required to ensure protection of these nearby facilities within 500 meters of the new well casing. If the well casing is connected to a non-metallic flowline, then bond all metallic pipelines, flowlines and trunklines that fall within the 500 meters of the new well to the well and together to ensure electrical continuity. Both air-cooled and oil-cooled rectifiers per 17-SAMSS-004 are permitted for well casing sites and these rectifiers shall be suitable for the location of installation.

6.5 Anodes and Anode Beds

6.5.1 Impressed current and galvanic anodes shall be manufactured in accordance with 17-SAMSS-007 and 17-SAMSS-006 respectively.

Note:

Nominal dimensional and weight data as specified in 17-SAMSS-007 for HSCI anodes are contained in Table 3 of this standard for ease of reference.

6.5.2 Anodes to be installed onshore deeper than 15 meters are classified as "deep" anode beds and as such require the drilling depth to be pre-approved in writing by the Saudi Aramco Groundwater Division, Reservoir Characterization Department. PMT shall obtain this approval.

6.5.3 Deep anode beds shall be installed in accordance with the latest revision of Saudi Aramco Standard Drawing AA-036385 "Cathodic Protection Deep Anode Bed".

Exception:

Where electrolyte resistivities or bore hole conditions are not suited to the conventional installation detailed on AA-036385, alternative material and installation configurations (unconventional deep anode bed) may be used. Add steel casings for the active columns of these unconventional anode beds to improve the anode bed performance in loss of circulation zones and include detailed calculations for the steel casing consumption.

6.5.4 For an anode bed discharging 25 amperes or more, a minimum distance of 150 meters shall be maintained between the nearest anode and the well casing to establish electrical remoteness.

Exceptions:

Where electrolyte resistivities at the anode bed are below 1,000 ohm-cm, the distance between the anode bed and the nearest well casing may be less than 150 meters, but must be far enough from the nearest well casing such that the calculated anodic gradient (ΔV) at the well casing is less than 1.0 volt using the resistivity determined at the anode bed.

For multiple well casing locations, the calculated difference in the anodic gradient between any two well casings must also be less than 200 mV.

6.5.5 For an anode bed discharging less than 25 amperes, a minimum distance of 75 meters shall be maintained between the nearest anode and the well casing.

Exceptions:

Where electrolyte resistivities at the anode bed are below 1,000 ohm-cm, the distance between the anode bed and the nearest well casing may be less than 75 meters, but must be far enough from the nearest well casing such that the calculated anodic gradient (ΔV) at the well casing is less than 1.0 volt using the resistivity determined at the anode bed.

For multiple well casing locations, the calculated difference in the anodic gradient between any two well casings must also be less than 200 mV.

6.5.6 Both mixed metal oxide (MMO) and high silicon cast iron (HSCI) anodes can be used for anode beds. Procure MMO and HSCI anodes from approved sources as specified by 17-SAMSS-007. Design parameters for HSCI and MMO impressed current anodes shall be as detailed in Table 2A and 17-SAMSS-007. Procure galvanic anodes from approved sources as specified by 17-SAMSS-006. The design parameters for galvanic anodes shall be as detailed in Table 2B, 17-SAMSS-006, and Saudi Aramco Standard Drawing AA-036389.

6.5.7 The HSCI impressed current anodes most commonly used by Saudi Aramco are listed in Table 3.

6.5.8 The ICCP current capacity of an ICCP anode bed shall be equal to or greater than the design current for the associated CP power source and shall be calculated as follows:

$$SA_{AB} \times I_{\varphi} \geq I_{\theta}$$

Where:

SA_{AB} = The total surface area of all the anodes in the anode bed

I_{φ} = Anode material current density per Table 2A

I_{θ} = CP power source rated current output

6.5.9 Adjacent ICCP anode beds powered from separate CP power sources shall be separated by a minimum distance of 50 meters. Adjacent ICCP anode beds powered from the same CP power source shall be separated by a minimum distance of 10.

6.5.10 Impressed current anodes shall be designed to maintain the minimum clearance detailed in Table 4 from any other cathodically protected structure.

6.5.11 Adjacent deep anode beds can be treated as individual anode beds if the separation between the anode beds meets or exceeds the minimum distances detailed in Table 4. Use actual measured soil resistivities if available.

Example:

Two 50 amp deep anode beds placed 75 meters apart in 2,500 ohm-cm soil can be installed 75 meters away from a buried pipeline.

6.6 Circuit Resistance

6.6.1 For a **T/R**, the CP system **rated** circuit resistance shall be defined as the T/R rated voltage, divided by the T/R rated current. Rated voltages and currents are as detailed on the manufacturer's data sheet/plate.
Example, 1 ohm is the rated resistance for a 50V/50A rectifier.

6.6.2 For a **photovoltaic** CP system, the **rated** circuit resistance shall be defined as the photovoltaic system rated output voltage divided by the CP power source minimum **required** rated current from Tables 1A, 1B, and 1C.

6.6.3 The CP system **operating** circuit resistance for an ICCP system shall be defined as the total effective resistance seen by the output terminals of the respective ICCP power supply, and for calculation purposes shall include:

a. Anode bed resistance-to-ground.

b. Positive cable resistance from CP power source to anodes.

c. Negative cable resistance from CP power source to structure(s).

d. Resistance of the casing to remote earth. This shall be as shown in Table 5 unless site testing is completed to verify a more accurate value.

e. Effective resistance caused by +0.8 volts anode bed back emf (for HSCI/MMO in coke breeze) plus the structure back emf per Table 5 (example: + 0.8 volts anode bed back emf -1.2 volts casing back emf = 2.0 volts total between the anode with coke breeze backfill and a coated steel casing).

R_{emf} = (+0.8V for anode bed back emf + structure back emf per Table 5) / I_{rated}

I_{rated} =CP power supply "rated" current

Exception:

A back emf of +1.0 volts (Cu/CuSO4) shall be used for MMO/HSCI anodes in seawater, Subkha, or comparable high salinity wet applications without coke breeze. This does not include the back emf associated with the polarized potential of the structure.

Note:

When using R_{emf} to calculate the maximum allowable anode bed resistance to provide 70% of R_{rated}, the value of R_{emf} must be multiplied by 0.7 (70%).

Example: $R_{anodebed}$ maximum = (0.7 x (R_{rated} - R_{emf})) - R_{cables} - $R_{structure}$.

For a 50 V / 50 Amp rectifier,

$$R_{anodebed}\ maximum = (0.7 \times (50/50 - (+0.8+1.2)/50)) - R_{cables} - R_{structure}$$

6.6.4 ICCP system designs shall take into consideration the calculated operating resistance and shall size the positive and negative cables and voltage rating of the T/R such that the "calculated" operating output of the T/R complies with all of the following:

a. At the design stage only, the target commissioning current shall be achieved at a voltage of lower than 70% of the T/R rated voltage output. A pre-commissioning report is approved if the well casing(s) is receiving the pre-commissioning current in the tables 1A, 1B, and 1C, rectifier is not exceeding 90% (±10%) voltage, anodes are not exceeding the pre-commissioning current and the rectifier current is compatible with the anodes while the system resistance does not exceed 90% and flowlines potential is meeting the criteria, otherwise send back to the design for a re-design and re-modeling.

b. The normal operating output current shall be achieved with the voltage adjustment set at more than 10% of the available (rated) T/R voltage output.

c. This is a mandatory design requirement but is not a mandatory requirement for commissioning acceptance.

6.6.5 The CP system "operating" circuit resistance measured during commissioning of a new CP system shall not be greater than 90% of the CP power supply "rated" circuit resistance unless all of the following are met:

a. If the CP system power supply is a tap adjustable T/R, it shall have at least two fine tap setting increments remaining.

b. Each anode current, for the required minimum number of anodes is at or below the maximum commissioning current rating (Table 3 of SAES-X-700).

Exception:

The system can discharge the minimum current required for commissioning listed in tables 1A, 1B, and 1C of SAES-X-700.

6.6.6 If during commissioning the flowline is measuring more than 5 Amps of current, then investigate if this is a bare flow lines and/or sacrificial anode drain, stray current drain, excessive grounding, large flowline network, major unknown changes to previous mechanical/electrical arrangements. Possible solutions include, to use only coated pipelines, identify changes to grounding and piping, verify P&ID drawings, include grounding to be part of CP design and simulation, isolate spools and consider resistance bonding, increase CP system rating to higher current (i.e., 100 amps instead of 50 amps) or carryout field investigation/study by PMT.

6.6.7 During commissioning, the compatibility of the anodes and rectifier rating is not mandatory if the commissioning current can be achieved.

6.6.8 The drilling contractor is held accountable for the functioning of the anode bed, must meet qualification requirements, either should have a NACE CP 2 qualified technician onboard or subcontract this service, prepare a clear procedure for anode bed installation with check and hold points and to be qualified by Saudi Aramco.

6.7 DC Power Supply

6.7.1 Cathodic protection rectifiers and photovoltaic systems shall be manufactured in accordance with 17-SAMSS-004 and 17-SAMSS-012 respectively.

6.7.2 The maximum allowed output rating for a DC power supply is 100 volts.

6.7.3 For hazardous areas (maximum Class 1 Zone 2), the design agency shall select a cathodic protection DC power supply (and other CP system equipment) that complies with the requirements of NEC Articles 500 to 504 for hazardous (classified) areas. CP equipment shall not be placed in Class 1 Zone 1 areas.

6.7.4 Rectifiers with NEMA Class 3R enclosures shall NOT be used inside hydrocarbon plant areas, within 30 meters of the plant perimeter fencing (outside), or within 1 km of a coastline.

6.8 Junction Boxes

6.8.1 Junction boxes shall be manufactured in accordance with 17-SAMSS-008.

6.8.2 Anode junction boxes used with DC power supplies with a rated output greater than 50 volts and a rated load resistance equal to or greater than
10 ohms shall utilize one of the following alternatives:

a. Shall use a non-metallic anode junction box, or

b. Shall have a protective clear plastic plate mounted in front of the shunts, with holes drilled to facilitate measurement of the current through each shunt.

6.8.3 Transformer/Rectifiers and junction boxes shall be permanently and systematically labeled in line with the well site electrical devices and equipment labeling system. Use T/R for transformer/rectifier and J/B for junction box.

6.9 DC Cables

6.9.1 Cathodic protection DC cables shall be manufactured in accordance with 17-SAMSS-017.

6.9.2 DC cables connected to a CP power supply either directly or through a junction box shall be shall be #6 (16 mm^2) or larger. Cables larger than
16 mm^2 shall be optimized in size to compliment the current capacity and resistance requirements of the respective CP system.

6.9.3 DC cables shall comply with the most recent edition of the National Fire Protection Association NFPA 70, National Electric Code (NEC).

6.9.4 Unless otherwise specified by the cable manufacturer, the allowable ampacity of High Molecular Weight Polyethylene (HMWPE) cables manufactured in accordance with 17-SAMSS-017 shall be rated for ampacity under the column for insulation rated for 90°C in the NFPA 70, NEC Handbook. Correction shall be made to an operating conductor temperature of 40°C.

6.10 Monitoring

6.10.1 The column titled “Monitor” in tables 1A, 1B, and 1C of this standard shall be used as a guideline for adjusting the output of cathodic protection systems to the optimum current output.

6.10.2 Remote monitoring

6.10.2.1 Where a remote monitoring system (RMS) is used, the design details as stated below for the RMS shall be submitted to the CP Proponent organization to review and confirm acceptance of the proposed RMS and their compatibility with the existing CP RMS.

a. A description of the existing CP RMS used by the respective CP Proponent in the respective area (if applicable). The description shall detail existing RMS hardware, software, communication connectivity and protocols.

b. Confirmation of proposed hardware and software compatibility with the existing CP RMS (if applicable), or written authorization by the CP Proponent organization to use a non-compatible RMS.

c. A data flowchart that illustrates the data path from the CP power supply to the CP Proponent’s desktop including connectivity details with software requirements and communication protocols.

6.10.2.2 Remote monitoring systems for well casing cathodic protection power supplies shall be designed and installed at all new CP system installations where an RTU is located within 500 meters of the CP power supply.

a. The CP RMS shall at minimum use signal transmitters for DC output voltage and current and a circuit breaker status switch.

b. The data collected and stored by the CP remote monitoring system shall be accessible by the CP Proponent organization directly from their desktop computer.

6.10.2.3 At locations where an existing useable RTU is not within 500 meters of the CP power supply or there is no RTU or there is no possibility of using the RTU, then a dedicated Remote Monitoring Unit shall be provided.

6.10.2.4 Remote Monitoring Units (RMUs) for cathodic protection power supplies shall comply with the requirements of 17-SAMSS-018.

6.10.2.5 Cables and field wiring used for the remote monitoring systems shall comply with SAES-P-104.

6.10.2.6 Signal Transmitter Requirements for CP Remote Monitoring Systems are as follows:

a. CP remote monitoring systems using signal transmitters shall utilize two (preferably loop powered) 4-20 mA signal transmitters to monitor the DC voltage and current. A third channel (digital) shall be used to monitor the CP power supply AC circuit breaker status.

b. The CP power supply shall be supplied with a circuit breaker containing a factory built-in auxiliary status switch to facilitate monitoring the CP power supply AC circuit breaker status. This type of circuit breaker must be clearly and specifically identified as a requirement on the Material Purchase Order.

c. The wiring between the circuit breaker auxiliary switch, signal transmitters, and the RTU must be sized such that the resistance is within the maximum load tolerance of the signal transmitter (typically 600 ohms maximum).

6.11 Bonding

6.11.1 Well casings shall be electrically continuous with their associated piping. Resistance bonding is not allowed during the design stage except to balance the current during commissioning as detailed below.

Exception:

If adequate current distribution is not achieved during commissioning or operation of a multi-well or shared cathodic protection system, the use of electrical isolation devices accompanied by a bond station may be added if approved by the CP Proponent organization. If resistors are necessary, they shall be welded tap adjustable or fixed (non-adjustable) resistors and shall be 0.15 ohms or less.

6.11.2 Electrically isolated flanged piping sections (spool pieces, i.e., Venturi spool) installed in a flow-line or trunk-line for use with instrumentation or other applications shall be bonded around using a metal bond strap fabricated to facilitate a reliable bond around the isolated equipment and ease of installation and removal, i.e., bolted.

Where a non-metallic pipeline or flowline is used, bond across to ensure the electrical continuity to all metallic spools, test lines, trunk lines using a dedicated bonding facility.

Bond all flowlines, test lines, trunk lines and pipelines through a dedicated bonding facility if they fall within 500 meters from a new well. Bond all crossing pipelines that fall within the 500 meters. For others farther away, bond at the closest pipelines crossing. If existing bonding is already present then no additional dedicated bonding is required. The proponent is responsible to monitor and restore the bonding stations and to upgrade the CP system to ensure adequate protection on all affected pipelines within the 1,000 meters. The bonding requirement stated here is to ensure that there is no stray current interference. Interference test shall be conducted during commissioning for all facilities that fall within 1000 meters to confirm that no stray current interference is present. Stray current interference if found shall be mitigated.

6.12 Electrical Isolation

Electrical isolation shall not be installed between a well casing and the associated flow-line.

Exceptions:

If adequate current distribution is not achieved during commissioning of a multi-well cathodic protection system, the use of electrical isolation devices accompanied by a bond station may be added after concurrence from the Supervisor of the CP Proponent organization and the design updated for further review & approval If resistors are necessary, they shall be welded tap adjustable or fixed (non-adjustable) resistors and shall be 0.15 ohms or less.

Where an electrical isolating gasket is used to isolate between two flange faces, it shall be installed at a location where the pipe is in a vertical orientation if practical.

7 Installation, Records, Commissioning, and Inspection

7.1 Where cathodic protection is required as specified in this standard, it shall be installed and pre-commissioned within one year after the drilling completion date of the well in all areas. If a rig works on an existing well; then, re-connection of the CP systems after well completion or drilling and workover work shall be done within 14 days from completing the work.

Exceptions:

1) *Installation and pre-commissioning of cathodic protection may be delayed until the flow-line has been installed at well casing locations where the well casing meets all of the following criteria:*
   a. *The well casing has not been externally coated or is a coated well drilled with a total vertical depth greater than 8,000 feet.*
   b. *AC power of distribution voltage is more than 5 km away.*
   c. *Soil or shallow aquifer (within 100 meter depth) resistivities are greater than 500 ohm-cm and not conducive to achieving full CP with galvanic anodes.*
2) *Installation of cathodic protection may be delayed for up to 2 years in Uthmaniyah and 3 years elsewhere for well casings where the associated flow-line or trunk-line has been installed and meets all of the following criteria:*
   a. *The trunk-line or flow-line has electrical continuity with the well casing through mechanical connection, or electrical bonding.*
   b. *The trunk-line is adequately cathodically protected.*
   c. *The flow-line or trunk-line is adequately protected at a measurement point 1 km away from the well casing.*
   d. *AC power of distribution voltage is more than 5 km away.*

7.2 All pre-commissioning and commissioning shall be done according to Tables 1A, 1B, and 1C of SAES-X-700, GI-0002.710, and SAEP-332.

7.3 Refer to Saudi Aramco Best Practice SABP-X-003 for detailed Installation, Records, and Inspection requirements. SABP-X-003 shall be deemed a mandatory document for this standard.

7.4 Refer to Saudi Aramco Engineering Procedure SAEP-332 for commissioning requirements. The criteria listed in this standard for commissioning are directly applicable to all pre-commissioning requirements.

Table 1: Table Title

Table 1A – CP Current Requirement for Gas Well Casings [(6)](#)

Gas Well Casings Cathodic Protection Current Requirements – Single Well								
Field Designation	**Bare Casing**				**Coated Casing**			
	Design		**Com-mission**	**Monitor**	**Design**		**Com-mission**	**Monitor**
	Minimum CP Power Supply Rating (amps)	Current Included for Flow-line (amps) (1,2)	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)	Minimum CP Power Supply Rating (amps)	Current Included for Flow-line (amps) (1,2)	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)
Abu Jifan, Fazran, Khurais, Mazalij, Midyan, Nuayyim, Shaybah, Dilam	25	3	20	15-20	10	3	2	1.5-5
Ghazal, Jufayn, Kassab, Manjurah, Sahba, Shaden, Tinat, Waqr, (ZMLH), (AWTD), (WDYH)	50	3	40	35-40	10	3	2	1.5-5
Haradh, Hawiyah, Harmaliyah, Midrikah, Nujayman, Shedgum, Uthmaniyah, (ANDR), (DAMM), (KRSN), Jafurah	75	3	40	35-40	75	3	35	30-35

Table 1B – CP Current Requirement for Water Injection and Oil Well Casings [(6)]

Oil Production and Water Injection Well Casings Cathodic Protection Current Requirements – Single Well								
	Bare Casing				**Coated Casing**			
	Design		**Com-mission**	**Monitor**	**Design**		**Com-mission**	**Monitor**
Field Designation	Minimum CP Power Supply Rating (amps)	Current Included For Flow-line (amps) [(1,2)]	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)	Minimum CP Power Supply Rating (amps)	Current Included For Flow-line (amps) [(1,2)]	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)
Uthmaniyah	50	3	35	30-35	15	3	7	5-7
Abqaiq, Abu Ali, Abu Hadriyah, AinDar, Berri, Dammam, Fadhili, Fazran, Hawiyah, Khursaniyah, Manifa, Qatif, Safaniyah, Shedgum	35	3	25	20-25	10	3	3	2-4
Ginah, Haradh, Hawtah, Harmaliyah, Midyan, Nuayyim, Shaybah, (ABMK), (Umjurf), (Hazmiyah), (Nislah), (Burmah), RGHB (Rughib), KHZM (Khuzama)	25	3	15	12-15	10	3	2	1-3
Abu Jifan, Khurais, Mazalij	10	3	4	2-5	5	3	1	0.5-5

Table 1C – CP Current Requirement for Water Supply Well Casings [3,4,5,6]

Water Supply Well Casings Deeper than the UER Formation Cathodic Protection Current Requirements – Single Well								
Field Designation	**Bare Casing**				**Coated Casing**			
	Design		**Com-mission**	**Monitor**	**Design**		**Com-mission**	**Monitor**
	Minimum CP Power Supply Rating (amps)	Current Included For Flow-line (amps) [1,2]	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)	Minimum CP Power Supply Rating (amps)	Current Included For Flow-line (amps) [1,2]	Minimum Casing Current (amps)	Optimum Casing Operating Current (amps)
Abu Jifan, Khurais, Mazalij,	10	3	2	1-3	10	3	0.5	0.2-0.5
All Other Fields	10	3	6	5-7	10	3	1	0.5-1

General Notes:

- *The following general notes are intended to address issues relating to tables 1A, 1B, and 1C.*
- *Fields indicated between two parentheses did not result from actual CPET logging, rather they are extrapolated based on their geological location*
- *Bare well casings are not recommended unless mandated by the proponent provided bare casings are allowed per the coating chart*
- *A tolerance of ±10% may be considered in the power source size in all tables 1A, 1B, and 1C current values*
- *Use 3 Amp of current for all flowlines at the design stage. Flowline current stated are for protection and not current drain on the flow lines. Appropriate current drain as required for flowlines and earthing shall be considered in the design as per field survey data.*
- *Verify the actual commissioning and monitoring currents using a clamp-on ammeter around the well casing*
- *PMT shall coordinate with Drilling and Workover to conduct CPET logging for new fields while inviting CSD to witness*
- *If the flowline is not constructed yet, then the pre-commissioning will be rejected unless a temporary cable is used to replicate the flowline.*
- *Flow lines and other pipelines that are not made of carbon steels, such as stainless steel, duplex stainless steel, etc., will require less negative CP potentials than that of the carbon steel pipelines in accordance with ISO-15589-1 & ISO-15589-2. As the protection criteria required for these pipelines are different from that of carbon steel pipelines, proper cathodic protection design consideration shall be made for the same to provide necessary protection whilst limiting the CP potentials to avoid hydrogen induced damages particularly for high strength materials.*
- *A pre-commissioning report is approved if the well casing(s) is receiving the pre-commissioning current in the tables, rectifier is not exceeding 90% (±10%) voltage, anodes are not exceeding the pre-commissioning current and the rectifier current is compatible with the anodes while the system resistance does not exceed 90% and flowlines potential is meeting the criteria, otherwise send back to the design for a re-design and re-modeling.*

- *It has been field demonstrated that a few gas fields such as UTMN, HWYH, SDGM and HRDH measured higher than normal current thru the F/L; therefore, a plan B shall be considered at the design stage to be prepared for a rectifier upgrade in case needed to satisfy the commissioning requirements.*

Table Notes:

(1) Flow line current stated are for protection and not current drain on the flow lines. CP current requirements for the flow-lines and trunk-lines of gas well casings shall be determined on a site specific basis. If additional current is necessary it shall be added to the current specified in tables 1A, 1B, and 1C. Include additional current for the grounding system.

(2) The current required for the flow-lines and trunk-lines is included in the "CP Power Supply Rating (amps)". If the flow-line or trunk-line is greater than 15 km long, additional current capacity requirements shall be determined through calculations completed in accordance with SAES-X-400. (3)Table 1C applies to water supply wells that extend through the wet UER formation such as typical Wasia water supply wells. Wells that do not extend below the UER (or other known corrosive formation), do not require CP. Wasia water wells drilled in areas where the UER is dry, do not require cathodic protection unless mandated by the proponent

(4) Regardless of the water well type, associated buried flow-lines must be cathodically protected in accordance with SAES-X-400.

(5) Water wells without cathodic protection, and within 500 meters of an impressed current CP system anode bed, or cathodically protected well casing must be bonded to the negative circuit of the CP system and supplied with sufficient CP current to ensure down-hole interference does not create an external corrosion problem.

(6) In this regard, the ΔV measured between the water well and the other protected structure within 500 meters should be less than 200 mV measured with extended lead wires, well head to protected structure.

**Table 2A – Impressed Current Anode
Consumption Rates and Maximum Design Current Densities**

Anode Material	Consumption Rate (kg/amp-y)	Anode Current Density (mA/cm^2)
High Silicon Cast Iron	0.45	0.7
Mixed Metal Oxide	Not applicable (see manufacturer's specification for 25 year life)	7.0

Commentary Note:

The consumption rates detailed in Table 2A include efficiency and the utilization factor for a HSCI tubular anode. Shortening of the effective anode length due to an end cap on the anode shall be neglected in the theoretical calculation of anode current capacity. Polymeric anodes can be used for flowlines or trunklines protection if conventional anodes have not been successful.

**Table 2B – Galvanic Anode
Consumption Rates and Open Circuit Potential**

Anode Material	Consumption Rate (kg/Amp-y)	Open Circuit Potential (*mV* vs $Cu/CuSO_4$)
Aluminum	3.8	-1,100
Magnesium	10.95	-1,700
Zinc	11.8	-1,100

Note:

The consumption rates detailed in Table 2B include efficiency; include 0.85 utilization factor in the calculation.

Table 3 – Impressed Current HSCI Anode Data

Type	Dimensions (nominal)	Weight (minimum)	Maximum Design Current Per Anode	Maximum Commission Current
TA-2	56 mm x 2133 mm	20.5 kg	2.63 amps	4.0 amps
TA-4	95 mm x 2133 mm	38.5 kg	4.45 amps	7.0 amps
TA-5A	121 mm x 2133 mm	79.0 kg	5.67 amps	10.0 amps

Note:

The maximum acceptable current during commissioning is based on manufacturer's ratings and shall not be used for design. This number shall only be used when determining commissioning acceptance. Anodes with output less than 1 Amp during commissioning should not be considered as acceptable. Tolerance values of ±10% during commissioning on the current is permitted in Table 3 above.

Table 4 – Minimum Distance to Nearest Cathodically Protected Structure

Anode Bed Rated Output Current (Amps)	Minimum Distance in Meters as a Function of Average Soil Resistivity at Anode Bed			
	ρ < 500 Ω-cm	500 Ω-cm ≤ ρ ≤ 1,000 Ω-cm	1,000 Ω-cm < ρ ≤ 3,000 Ω-cm	ρ > 3,000 Ω-cm
0 – 35	20 meters	25 meters	50 meters	75 meters
36 – 50	30 meters	35 meters	75 meters	150 meters
51 – 100	65 meters	75 meters	150 meters	250 meters
101 – 150	100 meters	125 meters	225 meters	350 meters
150+	Calculate the required distance to achieve an anodic gradient to remote earth of less than 1.8 volts using the soil resistivity measured at the anode bed.			

Exception:

The separation distance between an existing anode bed and a new buried structure shall be acceptable if field measurements on the buried structure at the nearest point to the anode bed demonstrate that the polarized (instant-off) potential is less than -1.20 volts (w.r.t. Cu/$CuSO_4$). Tolerance values of ±10% on all values are permitted in Table 4 above.

Notes:

1) *The distances detailed in Table 4 are provided to limit the structure polarized (instant-off) potential to less than -1.20 volts and to minimize the interference effects on other independent cathodically*

protected structures. Polarized potentials may be measured using a CP Assessment Probe or a CP potential coupon, as detailed in SAEP-333 Appendix A-2.

2) *For a new ICCP system, the "Anode Bed Output Current" value for Table 4 is the rated current output of the new CP power supply.*

Table 5 – Resistance of Casing-to-Remote Earth and Back EMF of Casings

Characteristic	Area	Bare Production or Water Injection Casing	Coated Production or Water Injection Casing	Bare Water Supply Well Casing	Flow-line / Trunk-line Network	Well Site Grounding Network (2)
Resistance of Casing-to Remote-Earth (ohms) (1)	Qatif	0.015	0.07	0.05	0.055	0.20
	Khurais	0.10	0.30	0.15	0.20	0.70
Structure Back EMF (volts) (1)	All Areas	1.17	1.2	1.15	1.2	1.0

Note 1: *The values contained in Table 5 are applicable to wells in the listed area. Areas with similar surface formation and casing completion characteristics can be assumed to be similar for design purposes. Otherwise, field testing is required in other areas to determine the appropriate design parameters for back emf and resistance to ground of the relevant structures. Field testing and/or modeling to determine the flowline/trunkline data in areas not listed above is required. For listed areas, field verification and/or modeling is recommended.*

Note 2: *Consideration of the well site grounding network especially gas sites is required where an extensive grounding network affects the current distribution at a well site such as would typically occur where down hole pumps are used, or where the field has been constructed with metallic power poles with an interconnected ground system.*

Table 6 - Recommended CP Rectifier Rating

Output DCV	Output DCA
30/50/75/100 V	5 A
30/50/75/100 V	10 A
30/50/75/100 V	15 A
30/50/75/100 V	25 A
30/50/75/100 V	35 A
30/50/75/100 V	50 A
30/50/75/100 V	75 A
30/50/75/100 V	100 A
30/50/75/100 V	150 A

Note:

For the purpose of well casing cathodic protection, rectifier ratings be selected from the recommended list above.

Appendices

Appendix 1 – Design Quality Assurance Check List

BI-____________JO-________________Well Number__________________

Item	Done	N/A	Details for the Proposed Anode Bed shown on CP Layout Dwg.
1	☐	☐	The number and type of anodes (TA-2, TA-4, or TA-5)
2	☐	☐	Proposed installation configuration, i.e., shallow or deep, and
3	☐	☐	if deep then confirmation of acceptable depth from Groundwater Protection Division
4	☐	☐	Diameter, depth, and number of holes and,
5	☐	☐	if more than one hole, the spacing of the holes
6	☐	☐	The distance between the oil/gas/water well and the proposed anode bed
7	☐	☐	The location of any buried pipeline within 200 meters of the proposed anode bed. If there are no buried pipelines (flow-line or other lines) within 200 meters of the proposed anode bed, please state this in the notes section of the drawing
Item	**Done**	**N/A**	**Details for Existing CP equipment shown on CP Layout Dwg.**
8	☐	☐	The age of existing anode bed and the number and type of anodes
9	☐	☐	The current distribution to existing wells for a multi-well system
10	☐	☐	Existing CP power supply operating output, or the measured operating resistance to verify correct current balance between new and existing
Item	**Done**	**N/A**	**Details for Proposed DC Power Supply shown on CP Layout Dwg.**
11	☐	☐	Rated DC volts and amps
12	☐	☐	Remote monitoring equipment details or a statement saying, “no remote monitoring”
13	☐	☐	The size and length of all DC cables for the proposed CP system
14	☐	☐	The CP junction boxes and their placement
Item	**Done**	**N/A**	**Calculations for Anode Beds (on CP Layout Dwg or on separate page)**
15	☐	☐	Maximum allowable CP system resistance
16	☐	☐	Maximum allowable anode bed resistance
17	☐	☐	Calculated anode bed resistance based on soil resistivity measurements taken at site
Item	**Done**	**N/A**	**Miscellaneous Requirements (on CP Layout Dwg or on separate page)**
18	☐	☐	Simulation sketch showing satisfactory current distribution to each well and associated flow lines & grounding
19	☐	☐	Verification statement that “ALL” buried ground cables on the site are insulated (jacketed)

Please provide remarks below for any item from above that is applicable but not done:

Item ___ ____

Item ___ ____

Designer’s Name (Print) _______________________Signature and Date ______________________

Designer’s NACE Cert # _________________________PMT Engineer Name (Print)_________________

Appendix 2 – Drill Stem and Test Anode Resistance Measurements

Well Designation:	____________	Submitted By:	________	Date:	______

New CP System Data ☐ **Place an X in the box if no new CP power supply or anodes**

CP Power Supply		Anode Hole(s)				Anodes		
DC Volts	DC Amps	Hole____ of ____	Target Resistance	Maximum Resistance	Max. Depth(m)	Quantity	What Type TA-2, 4 or 5	Other

Existing CP System Data ☐ **Place an X in the box if no existing CP power supply or anodes**

Existing CP Power Supply Rated Volts____ Rated Amps____			Existing Anode Hole(s)			Existing Anodes		
Operating DC Volts	Operating DC Amps	Manufacture Date	Shallow or Deep?	Number of Holes	Measured Resistance to Well Casing (Note 2)	Quantity	What Type TA-2, 4, or 5	Installation Year

Measurement Equipment Data:

Resistance Measurements	Length (meters)	Size (AWG or mm^2)	Resistance (ohms)
Resistance of the Test Wires and Connection (Note1)	25	Two wire #14AWG	
	200 (or more)	Two wire #14AWG	
Resistance of the Cable on the Test Anode (Note3)	100	16 mm^2	

Drill Stem and Test Anode Resistance Measurement Data (Report directly as read from meter)

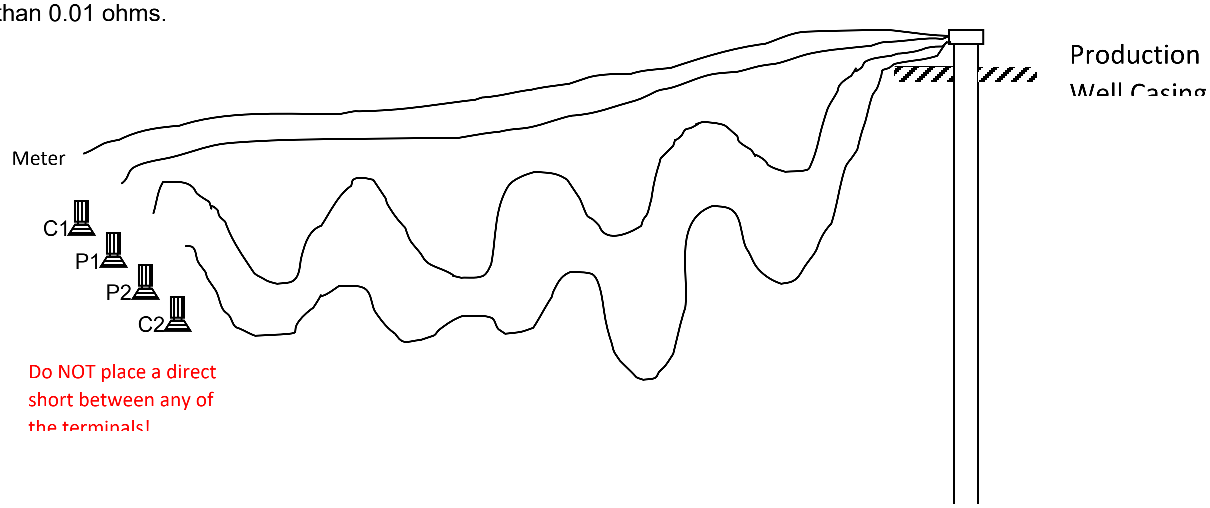
Depth (m)	Drill Stem(Note2) Resistance (ohms)	Test Anode(Note2) Resistance (ohms)	Soil Type	Remarks
Water level in the anode hole must be at the top of the casing before taking each measurement!!!!!!!				
6				
9				
12				
15				
18				
21				
24				
27				
30				
33				
36				
39				
42				
45				
48				
51				
54				
57				
60				
63				
66				
69				
72				
75				
Continued on next page.				

Appendix 2 – Drill Stem and Test Anode Resistance Measurements (continued)

Depth (m)	Drill Stem Resistance (ohms)	Test Anode Resistance (ohms)	Soil Type	Remarks
78				
81				
84				
87				
90				
93				
96				
99				
102				
105				
108				
111				
114				
117				
120				

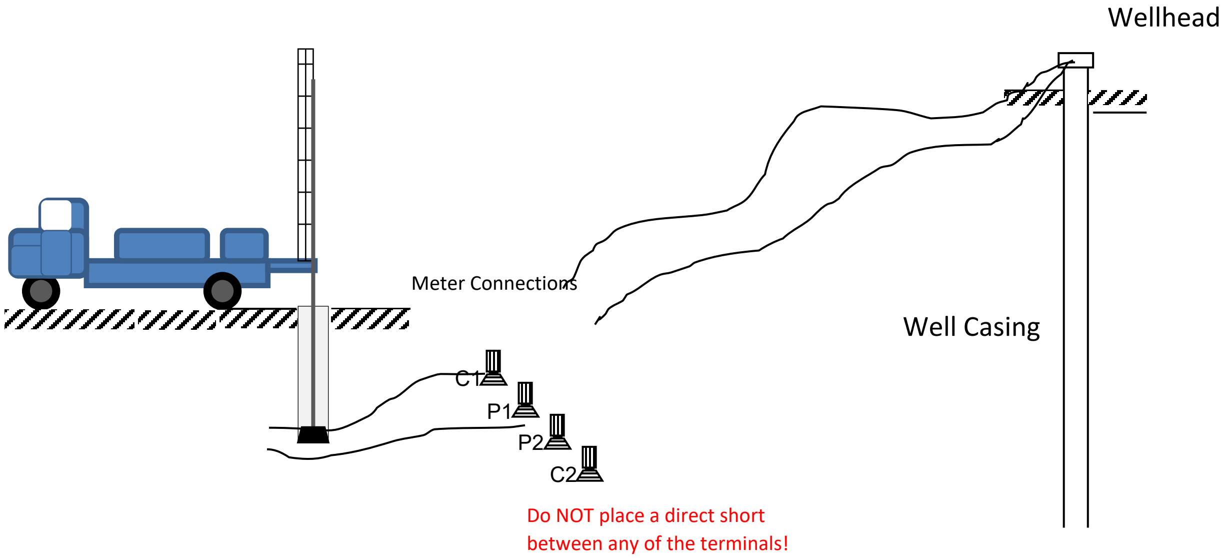
Notes:

1. **Procedure to measure the resistance of the "test wires and connections":**
   - Unspool 25 meters of #14 AWG wire. Connect one end to terminal C1 on the Megger.
   - Unspool another 25 meters of #14 AWG wire. Connect one end to terminal P1 on the Meter.
   - Tightly twist the open ends of the two 25 meter wires together and clamp to a bolt or flange on the well head (use vice grips or a "C" clamp – do not use alligator clip, spring clip or spring clamp).
   - Unspool 200 meters (typical) of #14 AWG. Connect one end to terminal C2 on the Meter.
   - Unspool another 200 meters (typical) of #14 AWG wire. Connect one end to terminal P2 on the Meter.
   - Tightly twist the open ends of the two 200 meter wires together and clamp to the same bolt or flange on the wellhead (use vice grips or a "C" clamp – do not use alligator clip, spring type clamp).
   - Measure the resistance with the Meter. Record the resistance measured as shown on the meter. It will typically be less than 0.01 ohms.



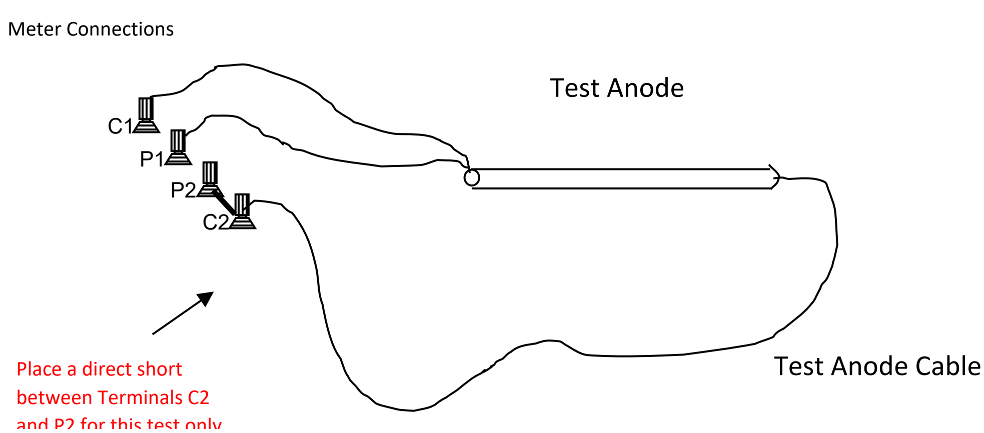
2. Procedure to measure the resistance of the drill stem, or test anode, or anode bed:

- Unspool 25 meters of #14 AWG wire. Connect one end to terminal C1 on the Meter.
- Unspool another 25 meters of #14 AWG wire. Connect one end to terminal P1 on the Meter.
- For Drill Stem , twist the open ends of the two 25 meter wires together and clamp to the frame of the drill truck (use vice grips or a “C” clamp – do not use alligator clip, spring clip or spring clamp).
  - For test anode resistance measurement, clamp these two wires to the end of the test anode cable instead of the drill truck.
  - For anode bed resistance measurement, clamp these two wires to the bus bar in the anode junction box instead of the drill truck.
- Unspool 200 meters (typical) of #14 AWG wire. Connect one end to terminal C2 on the Meter.
- Unspool another 200 meters (typical) of #14 AWG wire. Connect one end to terminal P2 on the Meter.
- Twist the open ends of the two 200 meter wires together and clamp to a bolt or flange on the wellhead (use vice grips or a “C” clamp – do not use alligator clip, spring clip or spring clamp)
- Record the resistance shown on the Meter.



3. Procedure to measure the resistance of the test anode cable:

- Unspool 25 meters of #14 AWG wire. Connect one end to terminal C1 on the Meter.
- Unspool 25 meters of #14 AWG wire. Connect one end to terminal P1 on the Meter.
- Twist the open ends of the two 25 meter wires together and clamp to one end of the anode (use vice grips or a “C” clamp – do not use alligator clip, spring clip or spring clamp).
- Uncoil the test anode cable and loop the cable end back towards the meter in a single loop.
- Short terminals C2 and P2 of the Meter (use a short strand of wire to connect terminal C2 to terminal P2).
- Connect the cable from the test anode to the P2 terminal.



Appendix 3 – Recommended Procedures

A3.1 Minimizing Casing Corrosion near Surface

To minimize corrosion of the well casings inside cellars, in the landing base area, or near surface, the following procedures should be implemented:

- Use an external FBE coating on the top two joints of all casings that extend to surface.
- Drilling operations should ensure that all cement is removed from the base of the cellar after each cement job to prevent the formation of a cement floor.
- Fill all annuli near surface with cement to displace air or water in the annuli.
- Seal the annular space between casings in the landing base area by seal welding the casings at surface (including the surface casing or conductor casing) to stop the ingress of air into the annular space between the casings.
- Seal (weld closed) any windows cut in the casings for top jobs, etc.

A3.2 External Coating on Well Casings

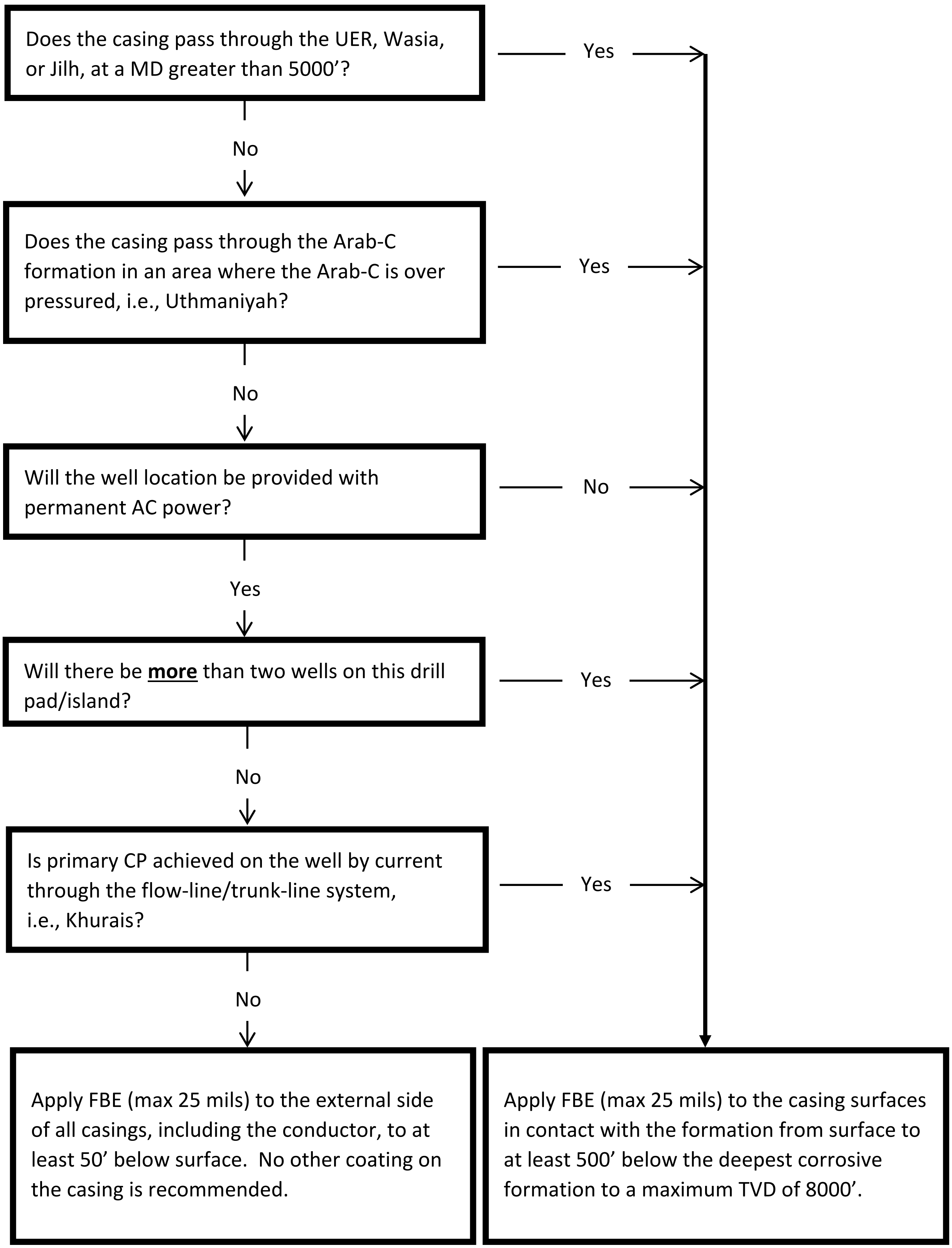
A3.2.1 The application of FBE to the external side of well casings may be used to effectively reduce the current requirement to approximately 10% of the cathodic protection current required for a similar non-coated casing.

A3.2.2 The application of FBE to the external side of well casings may be used to effectively increase the depth of influence of a cathodic protection system. Nominal attenuation characteristics for 16 mils (400 microns) of FBE coating applied to the external side of the well casing to a depth of 6,000 to 8,000 feet will typically extend adequate cathodic protection by 3,000 to 4,000 feet further into the bare casing section.

A3.2.3 The application of FBE to the external side of well casing tubulars should be restricted to a maximum thickness of 25 mils (625 microns).
Additional thickness creates problems due to the tight closing tolerances of the casing installation equipment.

A3.2.4 Casing with collar can be FBE coated effectively without removal of the collar if a preheat is completed on the collar before running the casing section through the assembly line induction oven, and by manually back spraying into the casing/collar crevice immediately before the collar enters the spray chamber.

A3.3 General Decision Chart for Coating "Onshore" Well Casings



A3.4 Determination of Well Casing Current Requirement

A3.4.1 Well casing corrosion in the Arabian Gulf area is typically caused by long line corrosion currents that are generated between two or more down-hole formations. The shallowest of the formations with an anodic response are typically more than 1000 feet deep and the area experiencing corrosion typically doesn't affect more than one or two lengths of casing.

A3.4.2 Corrosion can typically be attributed to poor quality cement in the casing/bore-hole annulus often amplified by the presence of flowing formation conditions or acid gases. Downhole corrosion cannot be detected by surface measurement techniques such as "E Log I", and cannot be reliably modeled using only formation resistivities and casing data.

A3.4.3 The most reliable method (and arguably the only reliable method) of determining the amount of CP current required to mitigate long line corrosion in a downhole formation is to run a CP evaluation log.
*(**Note:** CP evaluation logs are not effective for localized corrosion cells)*. Cathodic protection evaluation logs have been run by Saudi Aramco in the following fields; Abqaiq, Abu Ali, Ain Dar, Haradh, Hawiyah, Hawtah, Marjan, Nuayyim, Qatif, Safaniyah, Shaybah, Shedgum, and Uthmaniyah.

A3.4.4 Before CP is turned on; a cathodic protection evaluation log can be run to identify corrosive formations. However, due to the time delay for polarization of the casing, a log that is run to identify corrosion currents cannot reliably be immediately followed by a second log to determine the quantity of CP current required to overcome the corrosion currents.

A3.4.5 A CP evaluation log can be run to determine the amount of CP required to mitigate long line corrosion currents but a nominal amount of cathodic protection should be applied two weeks or more prior to running the log.

- When running the log, the actual amount of current going to the casing must be measured with a current clamp placed around the casing.
  It would be misleading to use the output current of the CP power supply because the flow-line and surface facilities associated with the rig will also take some of the current.
- The flow-line for the well will typically be physically disconnected during the logging procedures and must be bonded to the well head for the logging to accurately simulate actual operating conditions.
  This bond should be in place immediately after the flow-line is disconnected from the well head. If the bond is not installed, interference currents from nearby CP systems that would not occur when the flow-line is connected may occur when the flow-line is disconnected and in such cases would distort conclusions drawn from the logging information.
- For best results, the CP evaluation logging tool should be run at a maximum pull rate of 1,500 feet per hour and station stops should be made every 5 feet in areas where higher resolution is desirable, and every 10 feet in areas where lower resolution is acceptable.
- For best results, the cathodic protection logging tool should be left stationary at the lowest depth of the log for a period of 15 to

30 minutes before beginning the calibration part of the log. This delay is required to facilitate temperature stabilization of the tool.

- The cathodic protection evaluation logging tool should be provided with new blades at the beginning of each casing investigation, and should be provided with dielectric or electrically isolated stand-offs / centralizers. These are of particular importance for logging non-vertical casing segments.
- The use of the surface reference electrode (often termed “fish” by the logging vendor) is not required.

A3.5 Drilling Deep Anode Holes

A3.5.1 Drilling Equipment

Drill deep anode holes with rotary type drilling equipment that circulates the cuttings to surface by water (or water/mud mixture) through the hollow center of the drill pipe. Do NOT use equipment that drills a dry hole, or does not circulate drilling fluid to remove cuttings unless the formation characteristics dictate the use of a dry drilling technique such as is required in Khurais.

A3.5.2 Use of Casing

- Case the top 3 to 6 meters as required to maintain hole integrity through poorly consolidated surface soils.
- Do not install anodes inside any cased section of the hole.
- Use cement plugs to seal lost circulation areas, or bentonite if necessary. Do not use casing in areas where anodes will be placed (the casing will later corrode and the backfill dissipate into the formation leaving the anodes without complete backfill).
- If a water table is reached, casing can be installed to the top of the water table, but all anodes should be installed below the top of the water table to facilitate equalized current loading on the anodes.

Document History

23 January 2023	Major	Major periodical revision
19 November 2020	Editorial	Editorial revision to comply with SAEP-301 and SAEP-302
19 July 2020	Editorial	Editorial revision to change the contact name; to change the revision cycle from 3years to five years and to comply with SAEP-301 format
27 December 2017	Major	Major periodical revision
7 November 2013	Minor	Added clarification sketches to Appendix 2 (Drill Stem and Test Anode Analysis Procedure).
5 December 2012	Major	Major periodical revision